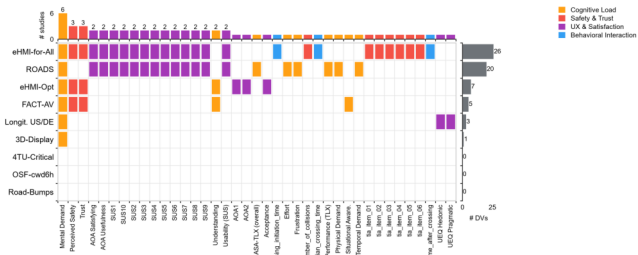


OpenDV-HCI: standardizing dependent variables turns fragmented open datasets into comparable, synthesizable evidence

(a) DV coverage across 9 datasets

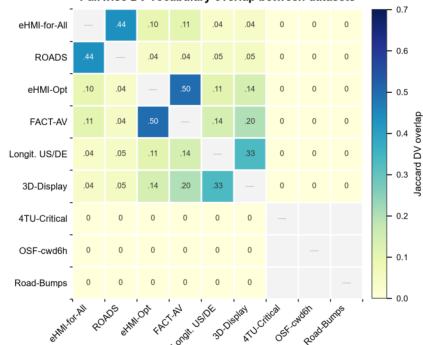
Dependent-variable coverage across 9 open HCI datasets



After schema-driven standardization, mean pairwise DV overlap is only Jaccard = 0.070. DVs spanning ≥ 3 studies: Mental Demand ($k=6$), Perceived Safety ($k=3$), Trust ($k=3$). 3 sensor/log datasets expose no canonical questionnaire DV.

(b) Pairwise overlap (Jaccard)

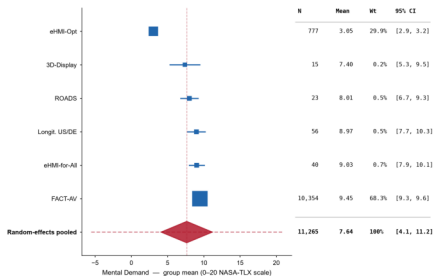
Pairwise DV-vocabulary overlap between datasets



(c) Meta-analysis: Mental Demand

Evidence synthesis unlocked by standardization: NASA-TLX Mental Demand

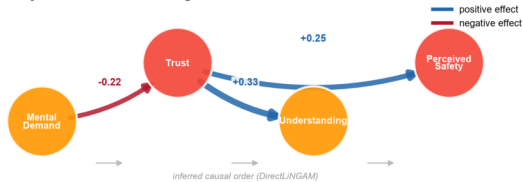
Pooling is only possible because heterogeneous source labels were mapped to one canonical DV. N is participant-level (within-subject rows auto-pooled to participant means). High P shows shared names can still mask scale/population differences.



Random-effects (DerSimonian-Laird), $k = 6$ studies. Heterogeneity: $I^2 = 99.9\%$, $\tau^2 = 19.21$, $Q(5) = 4001$, $p < 0.001$

(d) Cross-study causal structure

Cross-study causal structure among standardized DVs



DirectLINGAM on pooled within-study z-scores (complete-case, non-synthetic; $n = 10,354$ obs). Edge weights are structural coefficients.